

Frequency-specific sensory stimulation reduces stress potentiated deficits in 5XFAD mice

AUTHOR BLOCK: *T. C. FRANKLIN¹, H. MOORE¹, A. L. PAULSON², K. IYER³, A. SINGER³, M. GOODSON⁴, K. INAGAKI³, J. MATTA¹, A. PRICHARD⁶, R. DAS³, L. WOOD⁵, A. C. SINGER⁷

¹Biomed. Engin., Georgia Inst. of Technology- Emory Univ., Atlanta, GA; ²Biomed. Engin., Georgia Inst. of Technol., Cary, GA; ⁴Biomed. Engin., ⁵Inst. for Bioengineering and Biosci.,

³Georgia Inst. of Technol., Atlanta, GA; ⁶Res., Durham Veterans Affairs Med. Ctr., Durham, NC;

⁷Coulter Dept. of Biomed. Engin., Georgia Inst. of Technol. & Emory Univ., Atlanta, GA

Abstract:

Chronic stress substantially increases Alzheimer's disease (AD) risk, likely through stress-induced disruption of neuroimmune function and the gut microbiome. We and others previously demonstrated that noninvasive frequency-specific audiovisual stimulation ("flicker") modulates neuroimmune signaling in brain regions implicated in stress-related and neurodegenerative disorders. In unstressed mouse models of amyloidosis, 40Hz flicker improves microglial and neuronal function. Separately, we have shown that frequency-specific flicker attenuates stress-induced behavioral, synaptic, and inflammatory phenotypes in a sex-dependent manner. However, the combined impact of chronic stress and amyloid pathology, a clinically relevant multi-hit condition in which multiple risk factors converge to accelerate disease progression, remains poorly understood, and whether flicker can mitigate stress-potentiated deficits in this context is unknown. To address this gap, male and female 5XFAD mice were exposed to 4 weeks of chronic unpredictable stress (CUS) and concomitantly treated with 10Hz or 40Hz audiovisual flicker, frequencies previously shown to ameliorate stress- or amyloid-associated pathology in isolation. We assessed stress- and flicker-induced effects on behavior, amyloid pathology, microglia, and the gut microbiome. Amyloid deposition and microglial morphology were quantified in the hippocampus using 3D image reconstruction and analysis. CUS significantly exacerbated physiological stress responses in 5XFAD mice and produced locomotor and anxiety-like behavioral impairments. Flicker mitigated these deficits in a frequency- and sex-dependent manner, with the strongest protective effects observed in males. Chronic stress promoted a pathological trend characterized by greater amyloid burden, increased accumulation of larger plaques, and reduced microglial-amyloid interactions. Frequency-specific flicker prevented stress-associated plaque accumulation and restored microglial colocalization and engulfment of amyloid. Finally, 16S rRNA sequencing revealed sex-specific stress-associated alterations in the fecal microbiome that were mitigated by chronic flicker exposure. Together, these findings identify chronic stress as a potent amplifier of AD-related pathology and demonstrate that frequency-specific flicker counteracts behavioral, neuroimmune, amyloid, and microbiome dysfunction arising from the combined effects of stress and amyloidosis. These results support noninvasive flicker as a promising strategy for reducing neurodegenerative vulnerability in populations exposed to chronic stress.